

SMART INDIA HACKATHON 2026



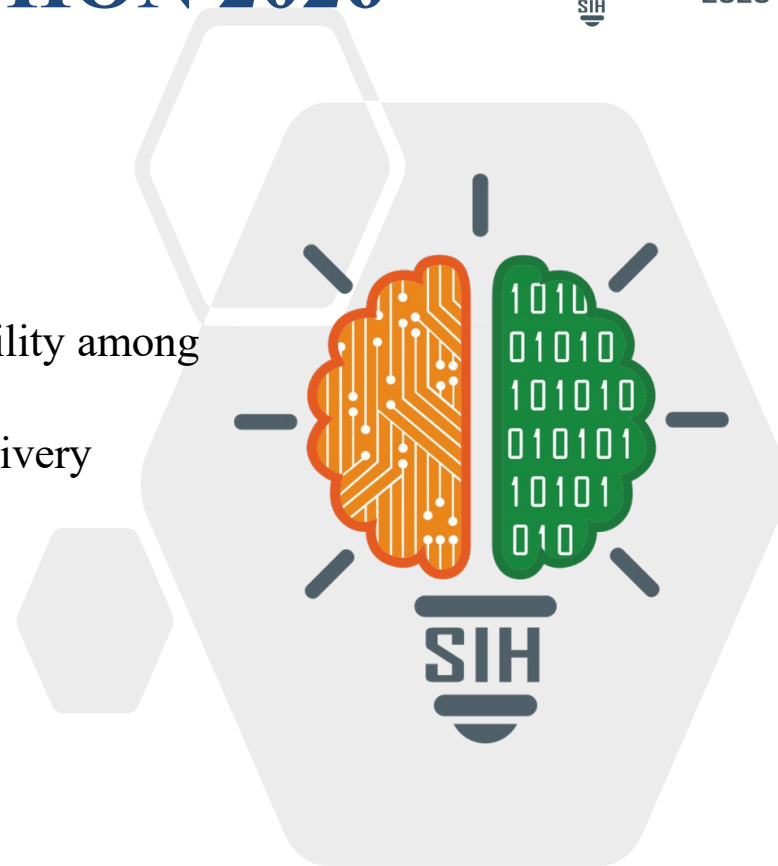
Problem Statement ID - SIH26129

Problem Statement Title - System integration and interoperability among government digital platforms, resulting in fragmented service delivery

Theme - Miscellaneous

PS Category – Software

Team Name – OneDesk



PROBLEM SOLUTION AND UNIQUENESS

SOLUTION

- **Federated Identity** — one OneDesk ID unifies every department's citizen record, while each department keeps its own data.
- **Consent-Based Exchange** — purpose-bound, time-limited, revocable access, modelled on India's DEPA / Account Aggregator standard.
- **Unified Application Tracking** — one live status across Identity, Business Registry, Tax and Municipal verification steps.
- **Universal Connector Framework** — REST, legacy SOAP and database systems join through adapters, no rewrite needed.
- **Operations Console** — real-time visibility into workflows, data quality, SLAs and full audit trails for officials.

ADDRESSAL APPROACH

Secure Data Exchange: A canonical field-level mapping layer resolves schema mismatches — e.g. full-name/DOB vs name/dateOfBirth — before any record crosses a department boundary.

Trust & Accountability: Purpose-bound consent tokens plus an immutable audit ledger mean access without valid consent is blocked outright, not logged after the fact.

Zero-Trust Security: Every service-to-service call is encrypted and mutually authenticated, with automated backups and alerting keeping the platform resilient by default.



Every cross-department request is **authenticated, signed and logged end-to-end** through Keycloak.

WHAT MAKES ONEDESK DIFFERENT

1

Keycloak-Powered Identity

One Keycloak realm mints every OneDesk ID and access token — no separate logins.

2

Zero Data Re-Entry

Submit once — departments reuse the same verified record instead of asking again.

3

Graceful Degradation

If a department's system goes down mid-workflow, only that step pauses and retries.

4

Progressive Modernization

Legacy systems join via adapters now and modernize later, on their own timeline.

5

Full Traceability

Every access and consent event is timestamped, for auditors and citizens alike.

ONE IDENTITY

ONE APPLICATION

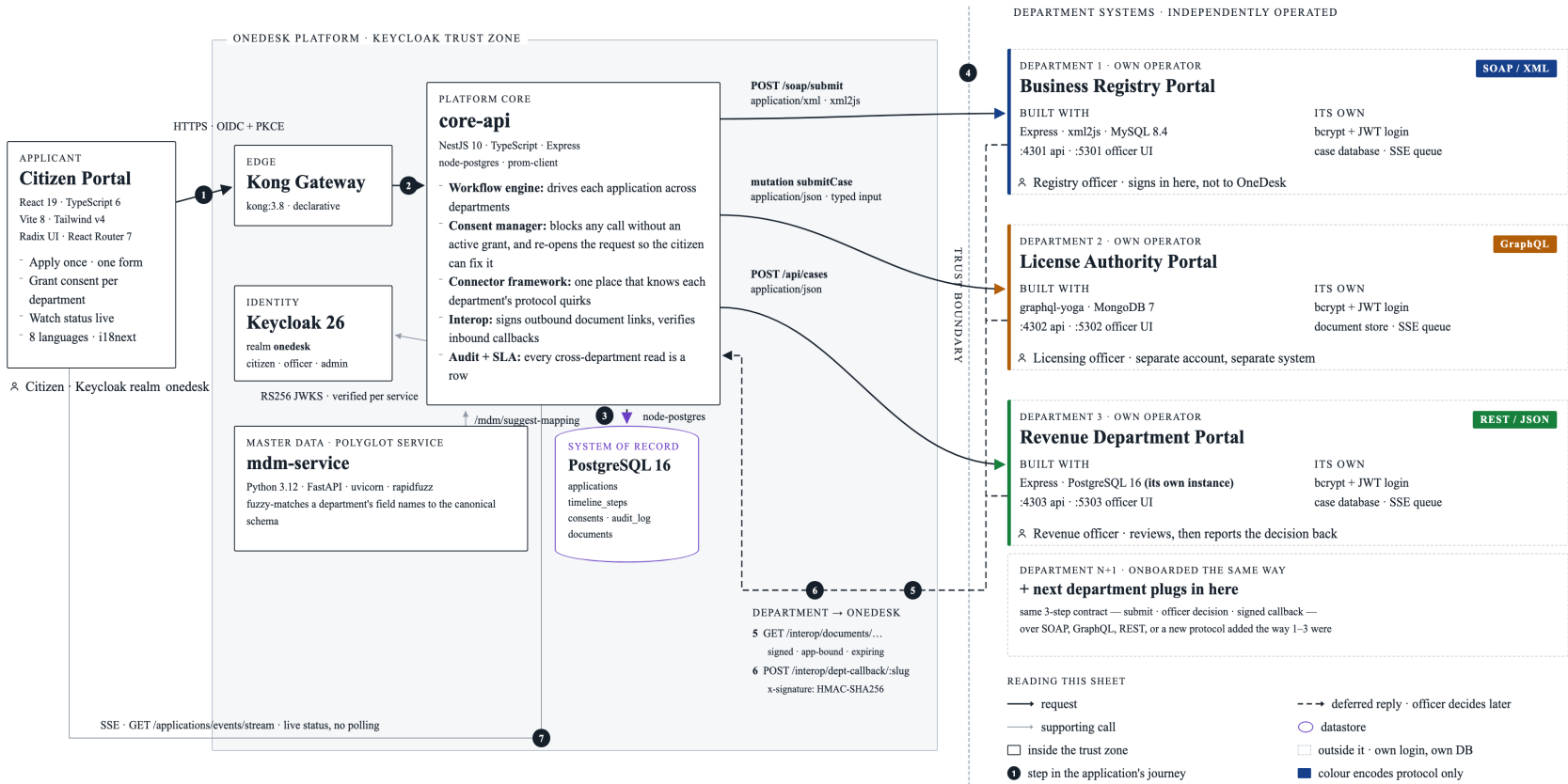
ONE PLATFORM

ONE TRUST

ONE EXPERIENCE

2

TECHNICAL APPROACH



VALIDATION RESULTS

3 PROTOCOLS

SOAP, GraphQL, REST — each exercised end-to-end, apply through certificate issued

4 DATABASES

Postgres x2, MySQL, MongoDB — proven interoperating, no shared schema

ZERO

SHARED CREDENTIALS

Each department's officer login is its own bcrypt+JWT, never Keycloak

STATELESS CORE-API

Behind Kong — add replicas, no sticky session, no in-memory workflow state

ASYNC

DEPARTMENT REVIEW

A slow department never blocks the platform — it waits in awaiting_department

TECH STACK & COMPONENTS · grouped by where each piece runs in the drawing above, not as a flat inventory

CITIZEN PORTAL

- React 19
- TypeScript 6
- Vite 8
- Tailwind v4
- Radix UI
- i18next

EDGE & IDENTITY

- Kong 3.8
- Keycloak 26

PLATFORM CORE

- Node.js
- NestJS 10
- Express
- PostgreSQL 16
- Prometheus

MASTER DATA

- Python 3.12
- FastAPI
- rapidfuzz

DEPARTMENT SYSTEMS

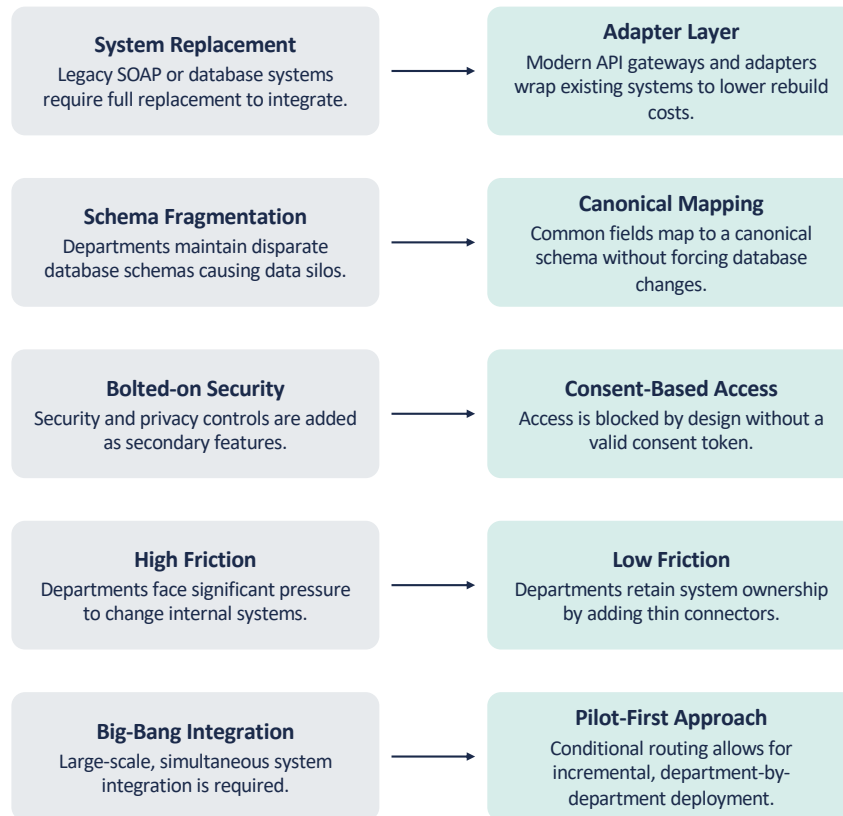
- SOAP / XML
- GraphQL
- REST / JSON
- MySQL 8.4
- MongoDB 7
- bcrypt + JWT

RUNTIME

- Docker Compose
- SSE
- HMAC-SHA256

FEASIBILITY AND VIABILITY ANALYSIS

IDEA FEASIBILITY



POTENTIAL CHALLENGES AND RISKS

- **Legacy & Scalability:** Old systems, inconsistent data, high traffic, and failures make integration difficult.
- **Trust & Adoption:** Departments may hesitate to share data, while citizens need clear consent and control.
- **Security & Privacy:** Requires encryption, access control, consent management, auditing, and secure data sharing.

STRATEGIES FOR OVERCOMING CHALLENGES

- **Legacy & Data Quality:** Standard adapters, data validation, and identity matching ensure accurate data.
- **Trust & Adoption:** Simple consent controls and small pilot programs build trust.
- **Security, Privacy & Resilience:** Zero-trust security, encryption, audit logs, backups, alerts, and automatic recovery ensure safe and reliable operation.

IMPACT AND BENEFITS

IMPACT

- Social**
Greater transparency, trust and accessibility.
- Economic**
Lower processing costs and delays.
- Governance**
Better coordination and accountability.
- Environmental**
Less paper and travel.
- Data Privacy**
Consent-based, secure data sharing.
- Scalability**
Extendable across departments and services

BENEFITS

- Citizens**
Faster, transparent, hassle-free services.
- Rural Citizens**
Less travel, paperwork no middlemen.
- Officers**
Less manual verification and faster processing
- Businesses**
Faster approvals and simplified licensing.
- Administrators**
Better monitoring and data-driven decisions.
- Vulnerable Groups**
Reduced time, travel and effort.



OneDesk adds an integration layer — existing departmental systems remain unchanged

STAKEHOLDER

Citizens / Users

- Easy service access & application tracking
- Less paperwork & office visits

Government Departments

- Secure data sharing & verification
- Less manual work & faster service delivery

Government Officials

- Efficient application, workflow & management
- Better monitoring & transparency

1

PLATFORM

Unified access for all government services

40-60%

Reduction in processing time

70% LESS

Paper usage and physical visits

HIGHER

Transparency, accountability & trust

CITIZEN-CENTRIC

Secure, consent-driven data architecture

RESEARCH AND REFERENCES

Government & Interoperability Frameworks

- [India API Setu](#) *MeitY · National interoperability platform*
- [India Enterprise Architecture \(IndEA\) 1.0](#) *Govt. of India enterprise framework*
- [India Stack](#) *UIDAI · NPCI · MeitY · Aadhaar, eKYC, UPI, DigiLocker*
- [X-Road Data Exchange Layer](#) *Estonia · national X-Road backbone*

4 SOURCES

Technical Standards & Integration

- [OAuth 2.0 Authorization Framework](#) *RFC 6749 · IETF*
- [OpenID Connect Core 1.0](#) *Identity layer over OAuth 2.0*
- [Canonical Data Model](#) *Enterprise Integration Patterns*
- [API Gateway Pattern](#) *microservices.io*

4 SOURCES

Research & Academic Papers

- [Sarantis et al. - Greek EGIF](#) *IEEE · one-stop service provision*
- [E-gov Data Interoperability, Hong Kong](#) *IEEE*
- [One-Stop-Shop e-Government Paradigm](#) *IEEE*
- [Cross-Border Interoperable Identity](#) *arXiv, 2023 · developing countries*

4 SOURCES

HOW RESEARCH SHAPED ONEDESK

Government Frameworks

API Setu · IndEA

Standards

OAuth2 · OIDC · X-Road

Research

IEEE · arXiv papers

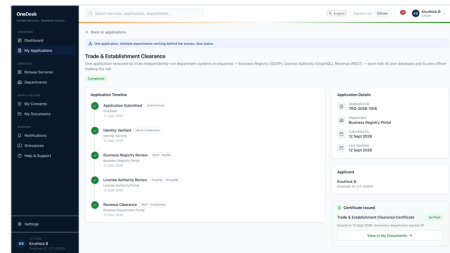
Architecture

Gateway · canonical model

OneDesk

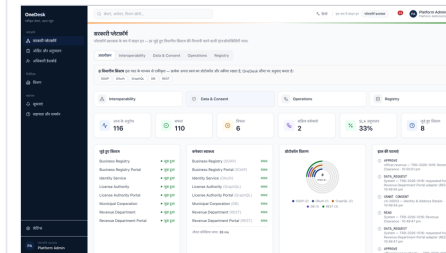
production system

Citizen Experience



- ✓ One application, tracked across three independent departments
- ✓ Live timeline - per-department status, no manual follow-up
- ✓ Consent-gated data sharing, visible to the citizen at every step
- ✓ Certificate issued automatically once every department signs off

Admin & Interoperability



- ✓ Single pane over every connected department system
- ✓ Per-protocol health - SOAP, GraphQL, REST, DB connectors
- ✓ Live protocol distribution and SLA compliance metrics
- ✓ Full audit trail of every cross-department data request

TEAM

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FUTURE WORK

→ Per-officer accounts and role-based access on department portals - today each portal runs on one shared seeded officer login

→ Production-grade deployment - orchestration and horizontal scaling beyond today's single-node Docker Compose setup

→ Wire the existing Prometheus metrics into a Grafana dashboard with SLA-breach alerting

GitHub : [One Desk - System Integrations and Interoperability among Govt Orgs](#)